

MATH 345 Skeleton Notes

Unit 7: Abstract vector spaces and polynomial approximation

Section 7.2: Inner product spaces

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Unit 4 used the dot product on $\mathbb{R}^n$ to define length, distance, orthogonality, projection, and least squares. An inner product space keeps exactly the rules needed for those ideas. This section introduces the abstract definition and keeps three examples in view: $\mathbb{R}^n$, polynomial spaces, and matrix spaces.

First, we define inner products.

Definition: Inner Products Let V be a vector space. An **inner product**

on V is a function $\langle \cdot, \cdot \rangle$ that assigns to each **ordered pair** of

vectors $\mathbf{v}, \mathbf{w}$ in V a **real number** $\langle \mathbf{v}, \mathbf{w} \rangle$ satisfying the following properties:

1. $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$ for all $\mathbf{v}, \mathbf{w}$ in V .
2. $\langle \mathbf{v} + \mathbf{w}, \mathbf{u} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V .
3. $\langle r\mathbf{v}, \mathbf{w} \rangle = r\langle \mathbf{v}, \mathbf{w} \rangle$ for all $\mathbf{v}$ and $\mathbf{w}$ in V and r in $\mathbb{R}$.
4. $\langle \mathbf{v}, \mathbf{v} \rangle > 0$ for all $\mathbf{v} \neq \mathbf{0}$ in V .

The vector space V together with $\langle \cdot, \cdot \rangle$ is called an

inner product space.

Theorem For an inner product $\langle \cdot, \cdot \rangle$ on a vector space V , the following additional properties automatically hold:

1. $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$ for any $\mathbf{u}, \mathbf{v}$ in V .
2. $\langle \mathbf{u}, r\mathbf{v} \rangle = r\langle \mathbf{u}, \mathbf{v} \rangle$ for any $\mathbf{u}, \mathbf{v}$ in V , and r in $\mathbb{R}$.
3. $\langle \mathbf{v}, \mathbf{0} \rangle = 0 = \langle \mathbf{0}, \mathbf{v} \rangle$
4. $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

The dot product on $\mathbb{R}^n$ is an inner product. We will often call the dot product the

standard inner product.

Activity

Let

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

be vectors in $\mathbb{R}^2$, and define

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 - u_2 v_1 - u_1 v_2 + 3u_2 v_2$$

Show this definition satisfies the 4th property of Theorem thm-7-4-inner-product-properties, i.e. that $\langle \mathbf{u}, \mathbf{u} \rangle > 0$ for $\mathbf{u} \neq \mathbf{0}$.

We calculate that

$$\begin{aligned}\langle \mathbf{u}, \mathbf{u} \rangle &= u_1 u_1 - u_2 u_1 - u_1 u_2 + 3u_2 u_2 \\ &= u_1^2 - 2u_1 u_2 + 3u_2^2 \\ &= u_1^2 - 2u_1 u_2 + u_2^2 + 2u_2^2 \\ &= (u_1 - u_2)^2 + 2u_2^2\end{aligned}$$

This quantity can only be zero if $u_1 - u_2 = 0$ and $u_2 = 0$. But this is only possible if $u_1 = 0$ and $u_2 = 0$, i.e. $\mathbf{u} = \mathbf{0}$.

Theorem Let $E = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ be the standard basis for $\mathbb{R}^n$, and let $\langle \cdot, \cdot \rangle$ be any inner product on $\mathbb{R}^n$. Let $c_{ij} = \langle \mathbf{e}_i, \mathbf{e}_j \rangle$ and consider the matrix $C = [c_{ij}]$. Then

1. C is a

symmetric matrix

.

2. $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T C \mathbf{v}$ for all $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$.

Definition Given an inner product $\langle \cdot, \cdot \rangle$ on $\mathbb{R}^n$, the symmetric matrix C analyzed in Theorem thm-7-4-inner-product-in-coordinates is called the

Gram matrix

 associated with the inner product.

Remark If a symmetric $n \times n$ matrix C satisfies $\mathbf{x}^T C \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$,

then C defines an inner product on $\mathbb{R}^n$. Moreover, these are the *only* symmetric matrices that define inner products. They are called *positive definite matrices*.

Definition: Positive Definite Matrices An $n \times n$ matrix A is

positive definite

if it is symmetric, and if $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq 0$.

Activity

Consider the inner product on $\mathbb{R}^2$ defined by

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 - u_2 v_1 - u_1 v_2 + 3u_2 v_2.$$

Find C such that $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T C \mathbf{v}$.

We employ Theorem thm-7-4-inner-product-in-coordinates. We compute that if $C = [c_{ij}]$, then

$$c_{11} = \langle \mathbf{e}_1, \mathbf{e}_1 \rangle = (1)(1) - (0)(1) - (1)(0) + 3(0)(0) = 1$$

$$c_{12} = \langle \mathbf{e}_1, \mathbf{e}_2 \rangle = (1)(0) - (0)(0) - (1)(1) + 3(0)(1) = -1$$

$$c_{21} = \langle \mathbf{e}_2, \mathbf{e}_1 \rangle = (0)(1) - (1)(1) - (0)(0) + 3(1)(0) = -1$$

$$c_{22} = \langle \mathbf{e}_2, \mathbf{e}_2 \rangle = (0)(0) - (1)(0) - (0)(1) + 3(1)(1) = 3$$

Therefore,

$$C = \begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}.$$

Definition: The Norm of a Vector In an inner product space, the

norm of a vector $\mathbf{v}$ is

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

The **distance** between two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

Definition: Unit Vectors A vector $\mathbf{v}$ in an inner product space V is

called a **unit vector** if $\|\mathbf{v}\| = 1$. This is equivalent to the equa-

tion $\langle \mathbf{v}, \mathbf{v} \rangle = 1$. The set of all unit vectors in V is called the **unit sphere** in V .

Theorem If $\mathbf{v}$ is a nonzero vector of an inner product space, then

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|} \mathbf{v}$$

is the unique unit vector that is a positive multiple of $\mathbf{v}$. We often say that $\mathbf{u}$ is the unit vector ‘in the same direction as $\mathbf{v}$ ’.

Definition: Orthogonal Vectors Two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner

product space are orthogonal if

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0.$$

The next examples show that inner products are not limited to ordinary coordinate vectors. The objects can be polynomials or matrices, as long as the rule for $\langle \cdot, \cdot \rangle$ satisfies the inner-product properties.

Spaces of polynomials

Activity

Let $V = \mathbf{P}_1$ be the vector space of all polynomials of degree at most 1. Define

$$\langle f, g \rangle = \int_0^1 f(x)g(x) dx.$$

Does this definition define an inner product on V ?

Let

$$f(x) = a_0 + a_1x \quad g(x) = b_0 + b_1x \quad h(x) = c_0 + c_1x$$

be arbitrary polynomials in $\mathbf{P}_1$.

We begin by checking that property (a) of the definition of an inner product holds, i.e. that $\langle f, g \rangle = \langle g, f \rangle$. Indeed,

$$\langle f, g \rangle = \int_0^1 f(x)g(x) \, dx = \int_0^1 g(x)f(x) \, dx = \langle g, f \rangle.$$

To verify that property (b) holds, we check that $\langle f + g, h \rangle = \langle f, h \rangle + \langle g, h \rangle$. We calculate that

$$\begin{aligned} \langle f + g, h \rangle &= \int_0^1 (f(x) + g(x))h(x) \, dx \\ &= \int_0^1 f(x)h(x) \, dx + \int_0^1 g(x)h(x) \, dx \\ &= \langle f, h \rangle + \langle g, h \rangle \end{aligned}$$

To verify that property (c) holds, we must check that for any scalar r , $\langle rf, g \rangle = r\langle f, g \rangle$. We calculate that

$$\langle rf, g \rangle = \int_0^1 (rf(x))g(x) \, dx = r \int_0^1 f(x)g(x) \, dx = r\langle f, g \rangle$$

To verify that property (d) holds, we must check that $\langle f, f \rangle > 0$ if $f \neq 0$. But

$$\begin{aligned}
\langle f, f \rangle &= \int_0^1 (a_0 + a_1 x)^2 dx \\
&= \int_0^1 (a_0^2 + 2a_0 a_1 x + a_1^2 x^2) dx \\
&= a_0^2 + a_0 a_1 + a_1^2/3 \\
&= (a_0 + a_1/2)^2 + a_1^2/12.
\end{aligned}$$

This quantity is positive unless $a_0 + a_1/2 = 0$ and $a_1 = 0$, and these equations only hold simultaneously when $a_0 = 0$ and $a_1 = 0$, i.e. when $f = 0$. Thus property (d) holds. Therefore the definition of the map $\langle, \rangle$ defined above is an inner product.

Activity

Consider P_1 with the inner product

$$\langle p, q \rangle = \int_0^1 p(x)q(x) dx.$$

Task Calculate the norm of the polynomial $p(x) = x$.

We calculate that

$$\begin{aligned}\|p\|^2 &= \langle p, p \rangle \\ &= \int_0^1 p(x)p(x) \, dx \\ &= \int_0^1 x^2 \, dx \\ &= \left[\frac{x^3}{3} \right]_0^1 \\ &= 1/3.\end{aligned}$$

Therefore, $\|p\| = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}$.

Task Calculate the distance between $q_1(x) = 2x + 1$ and $q_2(x) = 3x$.

We calculate that $q_1(x) - q_2(x) = (2x + 1) - 3x = -x + 1$, so

$$\begin{aligned}\|q_1 - q_2\|^2 &= \langle q_1 - q_2, q_1 - q_2 \rangle = \int_0^1 (1 - x)^2 dx \\ &= (-1/3)(1 - x)^3 dx \Big|_0^1 \\ &= 1/3.\end{aligned}$$

Therefore, $d(q_1, q_2) = \|q_1 - q_2\| = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}$.

Task Find the unit vector in the same direction as $p(x) = x$.

We calculated $\|p\| = \frac{1}{\sqrt{3}}$ in Task task-7-4-norm-practice. Therefore, the unit vector in the same direction as $p(x)$ is the polynomial

$$\begin{aligned}\frac{p(x)}{\|p\|} &= \frac{x}{\frac{1}{\sqrt{3}}} \\ &= \sqrt{3} \cdot x\end{aligned}$$

Task Are x and $x - 2/3$ orthogonal polynomials in P_1 ?

To determine if the two polynomials are orthogonal, we must check whether $\langle x, x - 2/3 \rangle = 0$. We calculate that

$$\begin{aligned}\langle x, x - \frac{2}{3} \rangle &= \int_0^1 x \left(x - \frac{2}{3} \right) dx \\ &= \int_0^1 \left(x^2 - \frac{2x}{3} \right) dx \\ &= \frac{x^3}{3} - \frac{x^2}{3} \Big|_0^1 \\ &= 0\end{aligned}$$

So the polynomials x and $x - 2/3$ are orthogonal to one another.

Spaces of matrices

Definition: The Trace of a Matrix The trace of a square $n \times n$ matrix A , denoted by $\text{Tr}(A)$, is the sum of its diagonal entries. For $A = [a_{ij}]$, we have

$$\text{Tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$

Theorem For any $n \times n$ matrices A and B ,

$$\operatorname{Tr}(A^T) = \operatorname{Tr}(A) \quad \text{and} \quad \operatorname{Tr}(A + B) = \operatorname{Tr}(A) + \operatorname{Tr}(B).$$

Activity

Consider $\mathbf{M}_{22}$, and let A, B be matrices in $\mathbf{M}_{22}$. Define

$$\langle A, B \rangle = \operatorname{Tr}(A^T B).$$

Task Find $\langle A, B \rangle$ for $A = \begin{bmatrix} 1 & 2 \\ 2 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 2 \\ -1 & 1 \end{bmatrix}$.

We calculate that

$$A^T B = \begin{bmatrix} 1 & 2 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ 3 & 1 \end{bmatrix}$$

Thus

$$\langle A, B \rangle = \operatorname{Tr}(A^T B) = -2 + 1 = -1.$$

Task Show that the map $\langle, \rangle$ defined above is an inner product.

We verify each of the four properties that define an inner product. Let A, B, C be arbitrary 2×2 matrices:

(a) We calculate that

$$\text{Tr}(A^T B) = \text{Tr}((A^T B)^T) = \text{Tr}(B^T (A^T)^T) = \text{Tr}(B^T A)$$

So $\langle A, B \rangle = \langle B, A \rangle$.

(b) We calculate that

$$\text{Tr}((A + C)^T B) = \text{Tr}(A^T B + C^T B) = \text{Tr}(A^T B) + \text{Tr}(C^T B)$$

So $\langle A + C, B \rangle = \langle A, B \rangle + \langle C, B \rangle$.

(c) We calculate that

$$\text{Tr}(r A^T B) = r \text{Tr}(A^T B).$$

So $\langle r A, B \rangle = r \langle A, B \rangle$.

(d) For $A \neq 0$ with $A = [a_{ij}]$,

$$A^T A = [c_{ij}] \text{ where } c_{ij} = \sum_k a_{ki} a_{kj}$$

The diagonal elements are: $c_{ii} = \sum_k a_{ki}^2 \geq 0$, and so

$$\text{Tr}(A^T A) = \sum_i \sum_k a_{ki}^2$$

is precisely the sum of the squares of the entries of A , which is always non-negative, and positive when at least one entry is non-zero. Thus $\langle A, A \rangle > 0$ for $A \neq 0$.

Task Find the distance between the two matrices

$$A = \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix}.$$

We recall that

$$d(A, B) = \|A - B\| = \sqrt{\langle A - B, A - B \rangle}.$$

We begin by calculating

$$A - B = \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ -1 & -3 \end{bmatrix}.$$

Next, we calculate that

$$(A - B)^T(A - B) = \begin{bmatrix} 4 & -1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ -1 & -3 \end{bmatrix} = \begin{bmatrix} 17 & 7 \\ 7 & 10 \end{bmatrix}.$$

Thus

$$\langle A - B, A - B \rangle = \text{Tr}((A - B)^T(A - B)) = 17 + 10 = 27.$$

Therefore, $d(A, B) = \sqrt{27} = 3\sqrt{3}$.

Task Find a value for x so that the two matrices

$$A = \begin{bmatrix} 1 & 0 \\ -2 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -3 & 2 \\ x & 0 \end{bmatrix}$$

are orthogonal.

We calculate that

$$A^T B = \begin{bmatrix} 1 & -2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ x & 0 \end{bmatrix} = \begin{bmatrix} -3 - 2x & 2 \\ 2x & 0 \end{bmatrix}.$$

So

$$\langle A, B \rangle = \text{Tr}(A^T B) = (-3 - 2x) + 0 = -3 - 2x = 0.$$

Now $\langle A, B \rangle = 0$ precisely when $x = -3/2$, and so this is precisely the value of x which makes the matrices A and B orthogonal.